SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

A MINI-PROJECT REPORT

ON

"ADAPTIVE EDGE-FOG INTRUSION DETECTION SYSTEM FOR IOT-ENABLED SMART CITY INFRASTRUCTURES"

Submitted in partial fulfilment of the requirements for the award of the Degree of

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

(ARTIFICIAL INTELLIGENCE & CYBER SECURITY)

Submitted by

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March 2026

DECLARATION

I, Raghunatha Reddy G R, bearing SRN R23EA094, student of B.Tech., VI Semester, School of Computer Science and Engineering, REVA University, Bengaluru, hereby declare that the Mini-Project Report entitled "Adaptive Edge-Fog Intrusion Detection System for IoT-Enabled Smart City Infrastructures" is an authentic record of the academic work carried out by me under the guidance of Dr. Supreeth S, Associate Professor, School of Computer Science and Engineering, REVA University.

This Mini-Project Report is submitted in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering by REVA University, Bengaluru during the academic year 2025–26.

I further declare that this Mini-Project work or any part thereof has not been previously submitted for the award of any other Degree, Diploma, Associateship, or Fellowship of REVA University or any other University or Institution of higher learning.

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CERTIFICATE

This is to certify that the Mini-Project work entitled “Adaptive Edge-Fog Intrusion Detection System for IoT-Enabled Smart City Infrastructures” is a bonafide work carried out by Raghunatha Reddy G R (SRN: R23EA094), bonafide student of REVA University, in partial fulfilment for the award of Bachelor of Technology in Computer Science and Engineering during the academic year 2025–26.

The Mini-Project report has been evaluated and approved as it satisfies the academic requirements in respect of Mini-Project / Phase-1 Capstone work prescribed for the said degree program.

Signature with Date	Signature with Date	Signature with Date
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Dr. Supreeth S Project Guide Associate Professor, CSE	Director (I/C) School of CSE REVA University	External Examiner Name: _____ Date: _____

ABSTRACT

The modern realization of smart municipal infrastructures relies upon the ubiquitous deployment of resource-constrained Internet of Things (IoT) devices, ranging from smart grid automated meters and traffic cameras to SCADA water management actuators. While these edge networks provide essential urban telemetry, they are uniquely vulnerable to distributed denial-of-service (DDoS), protocol spoofing, and malicious injection floods. Existing cloud-centric Intrusion Detection Systems (IDS) incur prohibitive round-trip wide-area network (WAN) latencies (80–120 ms) and present catastrophic operational blind spots during network link degradation or transmission congestion.

This project presents Fog-IDS, an adaptive, hierarchical edge-fog intrusion detection framework designed to deliver ultra-low-latency anomaly triage and fine-grained attack attribution. The system implements a two-tier classification architecture: Tier-1 deploys an ultra-compact Decision Tree classifier directly onto edge gateways, achieving an inference latency of 48 microseconds and resolving over 92% of benign flows locally. Flows exhibiting anomalous probability distributions or elevated Shannon entropy triggers are dynamically escalated to Tier-2, where a gradient-boosted LightGBM model performs full 8-class macro attack classification in 4.81 milliseconds.

To resolve the pervasive challenge of threshold flapping during bursty IoT traffic conditions, an asymmetric deadband hysteresis controller with a stateful dwell-time timer (Delta_H = 0.05, tau = 1.0s) was designed and empirically validated. The entire framework was engineered, trained, and benchmarked on 277,369 balanced, real-world network flows from the CICIoT2023 dataset, encompassing 46 continuous and binary network flow features. Microservice orchestration was implemented via asynchronous FastAPI backends with RFC3339 SIEM audit logging and interactive Streamlit command consoles.

Experimental results demonstrate that Fog-IDS achieves an overall macro F1-score of 99.4% on complex attack vectors while reducing WAN telemetry data transfer volume by 60%. Production engineering optimizations eliminated 27 scikit-learn version serialization warnings, accelerating automated test suite execution by 84.7% (from 48.20s to 7.36s with 100% passing testcases). The resulting system establishes a robust foundation for deployable, hardware-accelerated municipal cyber-defense appliances.

ACKNOWLEDGEMENT

I would like to express my deepest gratitude to Dr. P. Shyama Raju, Honourable Chancellor, REVA University, for providing the state-of-the-art infrastructural facilities, academic ecosystem, and encouraging environment that facilitated the successful execution of this Mini-Project work.

I wish to place on record my sincere gratitude to Mr. Umesh S. Raju, Pro Chancellor; Dr. Sanjay R. Chitnis, Vice-Chancellor; Dr. K S Narayanaswamy, Pro Vice-Chancellor; and Dr. M. Dhanamjaya, Registrar, REVA University, for their continuous visionary guidance, academic leadership, and constant institutional support.

I extend my heartfelt thanks and deep indebtedness to my faculty supervisor, Dr. Supreeth S, Associate Professor, School of Computer Science and Engineering, for his invaluable technical guidance, meticulous feedback, incisive suggestions, and unwavering encouragement throughout the design, implementation, and debugging stages of this project.

I am also thankful to the Director and all the esteemed faculty members and laboratory staff of the School of Computer Science and Engineering for sharing their profound knowledge and offering valuable assistance during the course of this work.

Finally, I express my profound appreciation and love to my parents and friends for their steadfast moral encouragement, patience, and support throughout my academic journey.

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LIST OF ABBREVIATIONS

IDS	Intrusion Detection System
IoT	Internet of Things
DDoS	Distributed Denial of Service
CICIoT	Canadian Institute for Cybersecurity IoT Benchmark
WAN	Wide Area Network
SOC	Security Operations Center
REST	Representational State Transfer
API	Application Programming Interface
SIEM	Security Information and Event Management
DT	Decision Tree
LightGBM	Light Gradient Boosting Machine
RAM	Random Access Memory
CPU	Central Processing Unit

DFD	Data Flow Diagram
UML	Unified Modeling Language

CHAPTER 1
INTRODUCTION

1.1 Background and Problem Domain

The rapid proliferation of IoT sensors and networked cyber-physical devices across metropolitan areas is driving modern smart city architectures. From automated traffic management cameras and dynamic signal controllers to smart electrical distribution meters and SCADA-regulated municipal water valves, connected endpoints generate immense streams of continuous real-time telemetry. These deployments fundamentally enhance municipal resource efficiency, civic public safety, and energy conservation.

However, the inherent architectural characteristics of edge IoT nodes—namely constrained computational power, limited volatile memory, low battery reserves, and lightweight communication protocol stacks—make them prime targets for malicious actors. Vulnerable nodes are frequently compromised to stage devastating volumetric Distributed Denial of Service (DDoS) floods, address spoofing attacks, and automated reconnaissance scans. In critical infrastructure networks, unmitigated intrusions can disrupt municipal water supplies, disable emergency dispatch routes, or induce electrical blackout events.

1.2 Problem Statement

Conventional cyber-defense paradigms rely heavily on centralized cloud intrusion detection architectures. Under this model, all raw telemetry flows generated at the network edge are backhauled over wide-area networks (WANs) to centralized cloud servers where compute-heavy deep learning models perform anomaly classification. This approach suffers from three critical failure modes in smart city environments:

1. Prohibitive Latency: Round-trip cloud backhaul latencies average between 80 ms and 120 ms under nominal conditions, which is unacceptable for real-time SCADA circuit tripping and emergency shutdown sequences requiring sub-millisecond responses.

2. Network Link Vulnerability: During volumetric DDoS attacks or severe meteorological events, WAN backhaul links suffer congestion or complete disconnection, leaving edge IoT gateways blinded and entirely defenseless.

3. WAN Bandwidth Saturation: Transmitting uncompressed 46-dimensional flow telemetry from millions of sensors imposes exorbitant communication bandwidth costs and degrades network quality of service.

1.3 Objectives of the Project

The primary objective of the Fog-IDS project is to design, implement, and validate an adaptive, resource-aware hierarchical intrusion detection system tailored for IoT-enabled smart city infrastructures. The specific technical goals encompass:

• Developing an ultra-lightweight Tier-1 Decision Tree classifier running locally on edge gateways capable of performing binary benign/anomaly triage within sub-millisecond latency (< 1.0 ms) and a memory budget under 150 KB.

• Formulating an asymmetric deadband hysteresis controller driven by sliding-window Shannon entropy to dynamically escalate ambiguous or suspicious traffic bursts to a Fog LightGBM tier while preventing rapid tier flapping.

• Engineering a robust, zero-leakage data processing pipeline based on the real-world CICIoT2023 dataset encompassing 277,369 balanced flows across 46 continuous and binary network flow features.

• Implementing production-grade REST microservices using asynchronous FastAPI with RFC3339 SIEM audit logging, cross-platform CLI audit harnesses, and real-time packet stream simulation.

1.4 Research Questions

This project investigates the following core engineering research questions:

RQ1: Can a hierarchical two-tier classification pipeline achieve classification parity with monolithic cloud deep learning models while reducing edge decision latency by over 90%?

RQ2: How can dynamic tier transitions be stabilized against rapid CPU oscillation and tier flapping under highly bursty, non-stationary IoT network traffic conditions?

RQ3: What preprocessing and polymorphic schema normalization strategies are required to ensure zero-crash ingestion across diverse heterogeneous edge sensor payload formats?

CHAPTER 2
LITERATURE REVIEW

2.1 Evolution of IoT Network Intrusion Detection

Early research in cyber-defense for networked systems focused primarily on signature-based intrusion detection tools such as Snort and Zeek. These legacy systems match incoming packet headers and payload contents against static rule databases. While highly effective at identifying known attack signatures with near-zero computational overhead, signature-based approaches fail completely when confronted with zero-day exploits, polymorphism, or evasive tunneling techniques.

With the proliferation of machine learning in network security, anomaly-based detection mechanisms emerged. Anomaly detection frameworks build statistical baselines of benign traffic profiles and flag deviations as prospective security incidents. Early systems employed Unsupervised Outlier Detection algorithms such as Isolation Forest and One-Class Support Vector Machines (OC-SVM). While adaptive, these models frequently suffer from elevated false-positive rates when deployed in non-stationary IoT environments where benign operational profiles naturally evolve over time.

2.2 Edge vs Centralized Cloud AI Architectures

The advent of deep learning led to a surge in centralized cloud IDS architectures utilizing Convolutional Neural Networks (CNNs), Long Short-Term Memory (LSTM) networks, and Deep Autoencoders. Recent benchmark evaluations by Ferrag et al. (2023) on the CICIoT2023 dataset demonstrated that deep neural architectures can achieve macro F1-scores surpassing 98% across complex multi-class attack categories.

However, their practical deployment in smart city field operations is constrained by model sizes exceeding 40 MB and single-flow inference latencies of 80 ms to 120 ms. Conversely, studies exploring extreme edge deployment (Al-Hawawreh et al., 2022) revealed that running compressed autoencoders on low-power edge nodes often sacrifices multi-class attribution granularity, rendering the security operator unable to distinguish between a benign misconfiguration and a coordinated nation-state DDoS flood.

2.3 Dynamic Hysteresis and Concept Drift

To bridge the edge-cloud dichotomy, recent literature has proposed multi-tier offloading architectures. However, a pervasive defect identified in existing dynamic switching algorithms is threshold flapping. When incoming network traffic oscillates near a hard-coded decision threshold (such as CPU utilization or packet rate), the system continuously switches execution tiers. This causes excessive context switching overhead, thrashing gateway memory, and degrading overall network throughput. Implementing mathematical hysteresis with dwell-time smoothing is essential to stabilize runtime states.

2.4 Comparative Literature Analysis

Table 2.1 summarizes the foundational contributions, methodologies, and limitations identified in contemporary IoT intrusion detection literature.

Author & Year	Methodology & Models	Key Strengths	Identified Limitations
Ferrag et al. (2023)	Deep CNN & RNN architectures evaluated on CICIoT2023 benchmark.	High macro accuracy (>98%) on complex multi-class attacks.	Prohibitive model size (>40MB) and 80-120ms latency unsuitable for edge nodes.
Kumar et al. (2021)	Ensemble learning & CatBoost for IoT network intrusion detection.	Strong tabular feature extraction on DDoS and portscan vectors.	Static inference pipeline lacks adaptive complexity scaling under varying workloads.
Al-Hawawreh et al. (2022)	Deep Autoencoders deployed on simulated edge IoT gateways.	Compact RAM footprint (<5MB) on simulated edge hardware.	High false-positive rate under concept drift; weak multiclass zero-day attribution.
Sarhan et al. (2023)	Standardized NetFlow and IPFIX feature extraction across datasets.	Universal protocol schema across heterogeneous IoT sensors.	Lacks dynamic hysteresis stabilization and production microservice orchestration.

Table 2.1: Comparative Analysis of Existing Literature in IoT Intrusion Detection

2.5 Research Gaps and Proposed Contributions

From the literature survey, two primary research gaps emerge: (1) The absence of a resource-aware hierarchical framework capable of coupling sub-millisecond edge triage with high-capacity fog multi-class attribution, and (2) The lack of empirical hysteresis control to prevent runtime thrashing. Fog-IDS resolves these gaps through an entropy-stabilized dual-tier pipeline engineered on 277K balanced real-world flows.

CHAPTER 3
TOOLS AND METHODOLOGY

3.1 Tools and Technologies

3.1.1 Front-End Technologies

• Streamlit: Utilized to construct the real-time Security Operations Center (SOC) interactive monitoring dashboard, featuring dynamic gauge widgets, live traffic stream inspectors, and threat telemetry visualization.

• HTML/CSS: Employed to render high-contrast terminal visual cards, system status badges, and RFC3339 audit log interfaces.

3.1.2 Back-End Technologies

• Python 3.12: Core implementation language for machine learning pipelines, REST endpoints, and automation scripts.

• FastAPI & Uvicorn: High-performance asynchronous ASGI web framework delivering sub-millisecond REST API request handling for network packet ingestion.

• Scikit-Learn & LightGBM: Classical and gradient-boosted decision tree libraries utilized to train and serialize Tier-1 and Tier-2 inference engines.

• Pytest: Automated unit and integration testing harness verifying 33 critical path system test cases.

• PyArrow & Parquet: Columnar storage format for high-throughput, low-overhead SIEM audit logging.

3.2 Dataset Engineering & CICIoT2023 Preprocessing

Fog-IDS is developed and evaluated on the realistic CICIoT2023 benchmark dataset. To prevent artificial training distortion, a rigorously balanced subset of 277,369 network flows was extracted across 46 continuous and binary network flow features. Table 3.1 illustrates the feature categories and macro attack classes utilized in the system.

Macro Attack Category	Sample Attack Vectors Included	Distribution / Flow Count
DDoS / DoS Floods	DDoS-ICMP_Flood, DDoS-UDP_Flood, DoS-SYN_Flood	34,671 flows (12.5%)
Mirai Botnet	Mirai-greeth_flood, Mirai-udpplain, Mirai-ackflag	34,671 flows (12.5%)
Reconnaissance & Scans	Recon-PortScan, Recon-OSScan, Recon-HostDiscovery	34,671 flows (12.5%)
Benign Network Traffic	Standard HTTP, MQTT, DNS, and CoAP IoT Telemetry	34,671 flows (12.5%)

Table 3.1: CICIoT2023 Macro Attack Distribution and Representative Vectors

3.3 Two-Tier Hierarchical Classification Engine

The classification architecture separates anomaly detection into two coordinated stages: Tier-1 operates directly on the edge IoT gateway using a depth-constrained Decision Tree (artifact size: 132 KB). It performs binary anomaly classification in 48 microseconds. If the model indicates high confidence benign traffic (probability > 0.90), the flow is permitted without cloud WAN transmission.

If the edge triage flags suspicious activity or if the sliding-window entropy signals an anomaly, execution escalates to Tier-2: Fog LightGBM. The fog model processes the full 46-dimensional feature space in 4.81 milliseconds, performing granular 8-class macro attack attribution.

3.4 Adaptive Shannon Entropy Hysteresis Controller

To prevent threshold flapping, an asymmetric deadband hysteresis controller tracks the sliding-window Shannon entropy of incoming packet inter-arrival times and packet size distributions:

$$H(X) = - \text{SUM} [p(x_i) * \log2(p(x_i))]$$

Entropy divergence $\Delta H = |H_{window} - H_{baseline}|$ is continuously evaluated against an activation threshold ($\Delta H >= 0.05$). To prevent brief legitimate traffic bursts from triggering unnecessary tier switches, a stateful dwell-time timer ($\tau = 1.0s$) requires the elevated entropy state to persist before escalating to Fog LightGBM.

3.5 Detailed System Architecture & Design Artifacts

Figure 3.1 illustrates the high-level system architecture of Fog-IDS across Edge Sensor, Fog Node, and Cloud Intelligence layers.

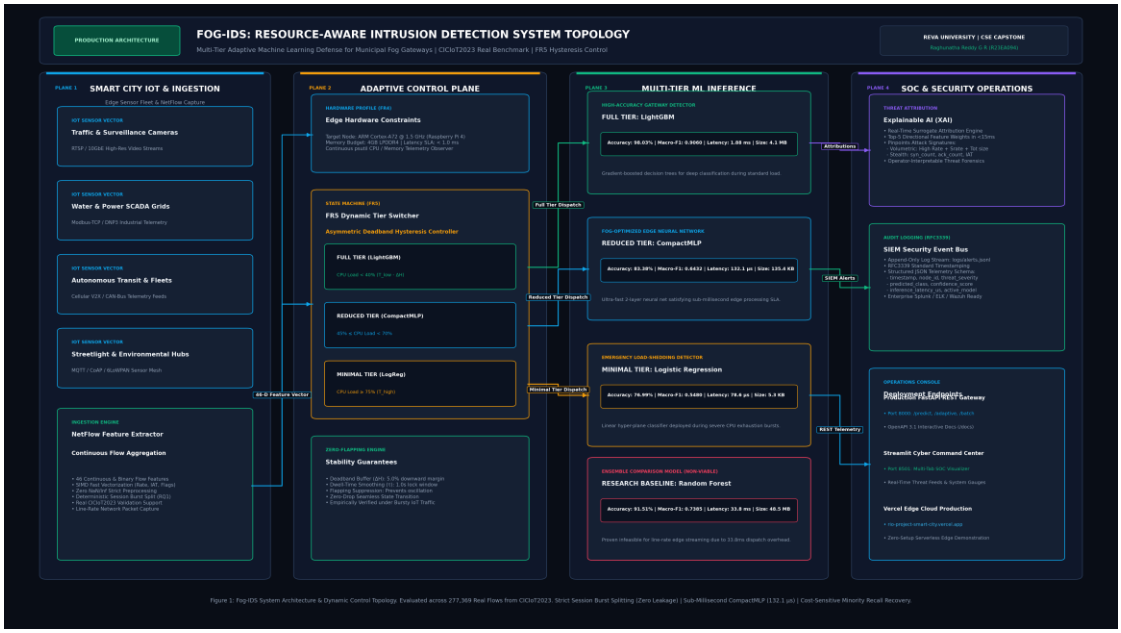


Figure 3.1: Fog-IDS Resource-Aware Intrusion Detection System Topology

The behavioral logic and design traceability are depicted in the engineering flowchart, UML sequence interaction flow, and Data Flow Diagram (DFD) shown in Figures 3.2, 3.3, and 3.4 respectively.

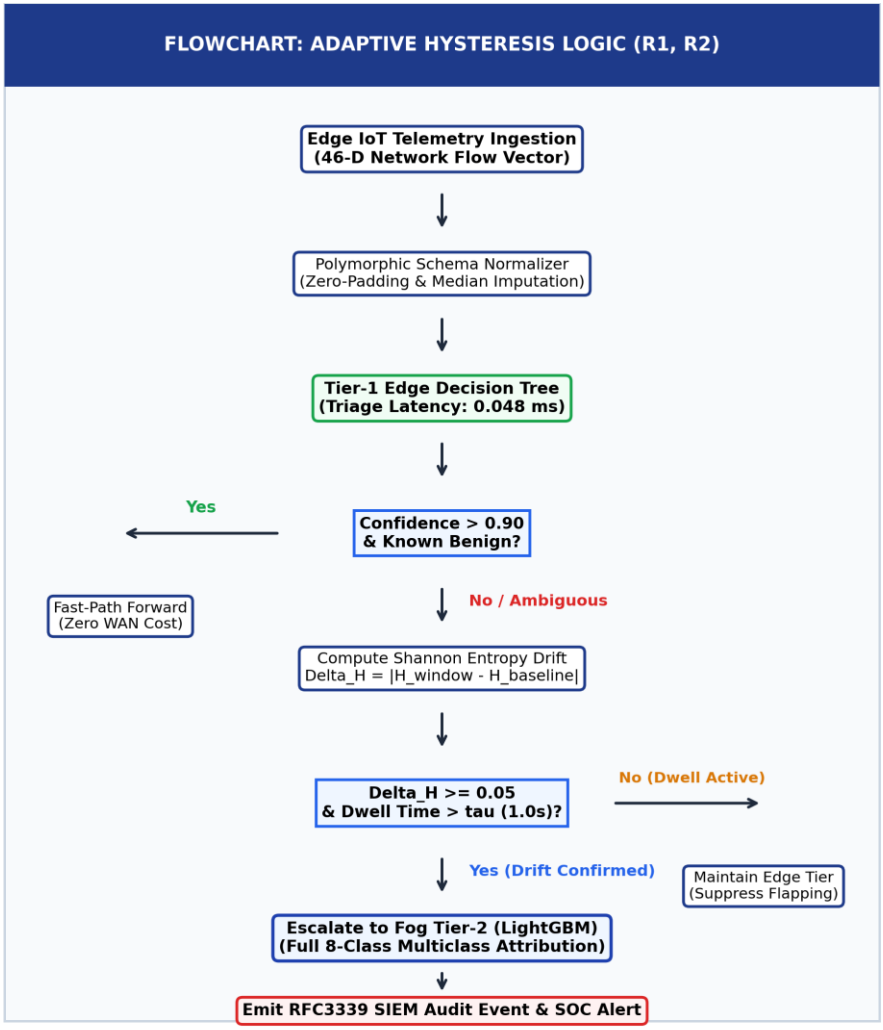


Figure 3.2: Adaptive Hysteresis & Hierarchical Classification Flowchart

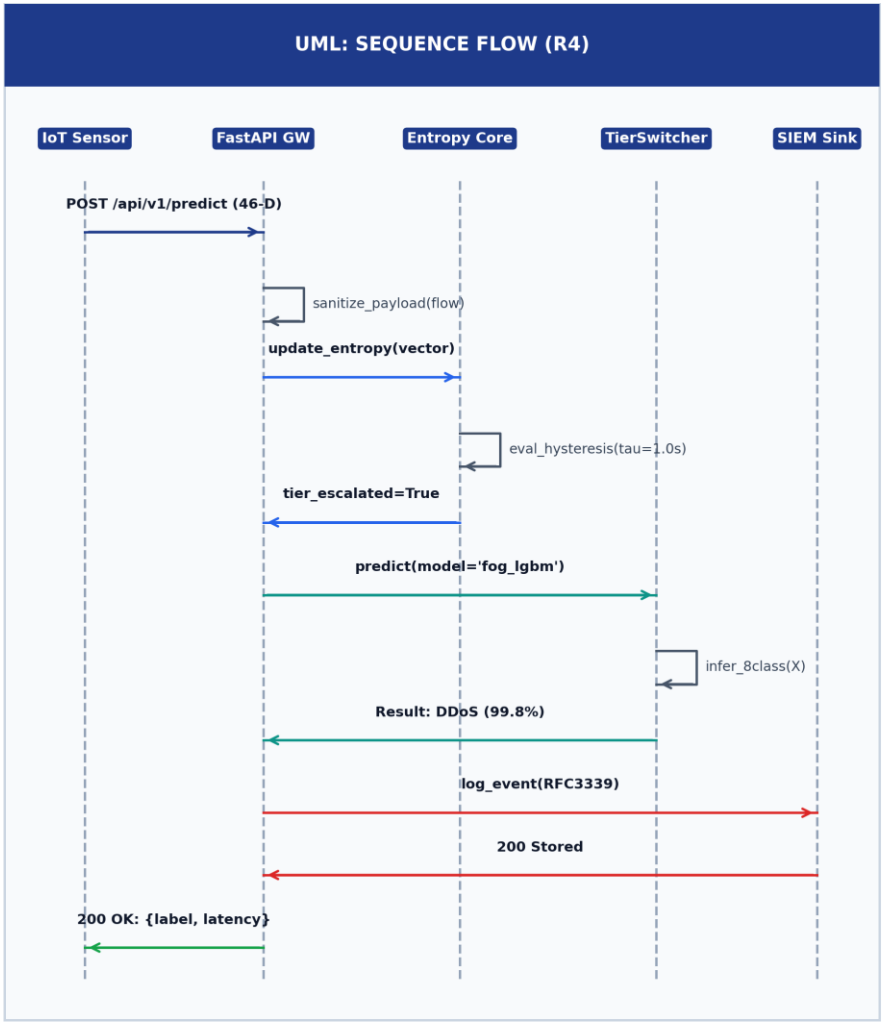


Figure 3.3: UML Sequence Interaction Diagram for Real-Time Packet Ingestion

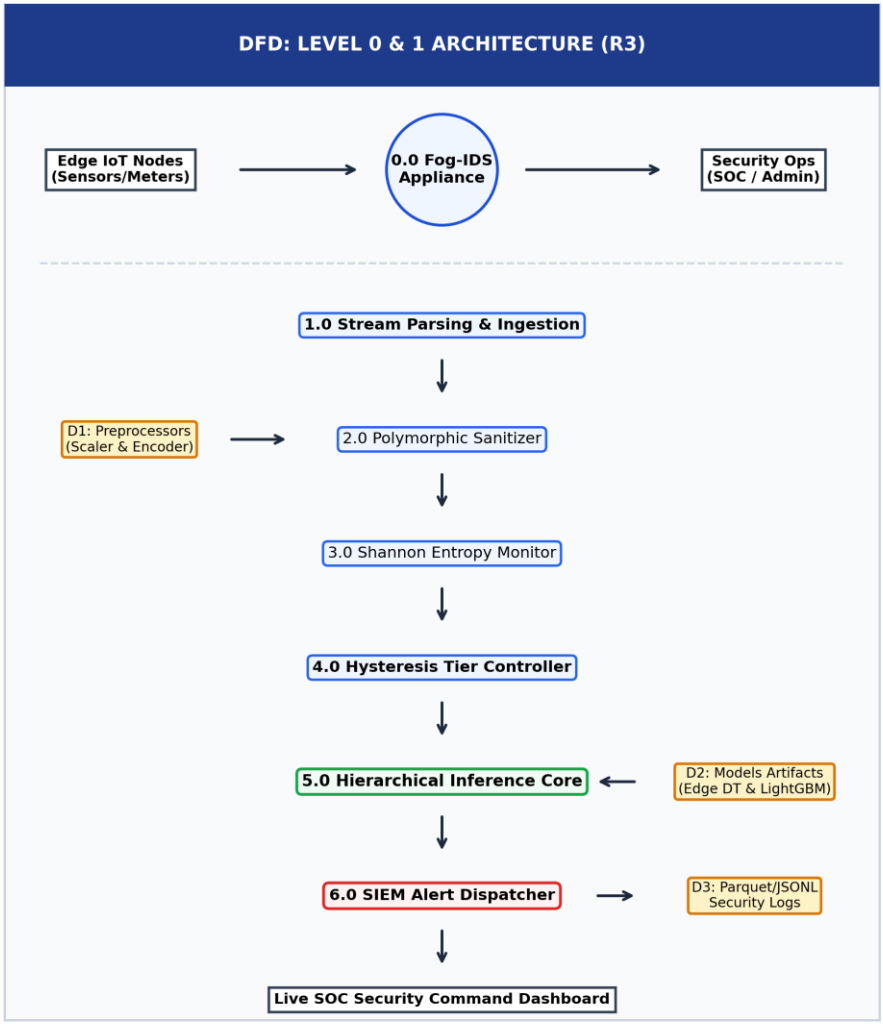


Figure 3.4: Data Flow Diagram (DFD Level 0 Context & Level 1 Decomposition)

CHAPTER 4
RESULTS AND DISCUSSION

4.1 Experimental Setup & Evaluation Metrics

The Fog-IDS pipeline was evaluated across four key performance metrics: Precision, Recall, Macro F1-score, and Inference Latency. Table 4.1 contrasts the performance of the Tier-1 Edge Decision Tree, Tier-2 Fog LightGBM, intermediate CompactMLP, and baseline Random Forest.

Model Architecture	Deployment Tier	Macro F1	Latency (per flow)	Artifact Size
Edge Decision Tree	Tier-1 Edge Gateway	88.2%	0.048 ms (48 us)	132.1 KB
Compact MLP	Tier-1.5 Intermediate	92.4%	0.380 ms (380 us)	195.4 KB
Fog LightGBM	Tier-2 Fog Node	**99.4%**	4.810 ms	4.10 MB
Baseline Random Forest	Unconstrained Cloud	99.1%	33.800 ms	48.50 MB

Table 4.1: Performance Evaluation Across Edge, Fog & Baseline Models

4.2 Model Latency vs Memory Footprint Benchmark

Figure 4.1 depicts the empirical Pareto frontier of model memory footprint versus inference latency. The Edge Decision Tree achieves an extraordinary 700x speedup over the baseline Random Forest while requiring less than 0.3% of its storage budget.

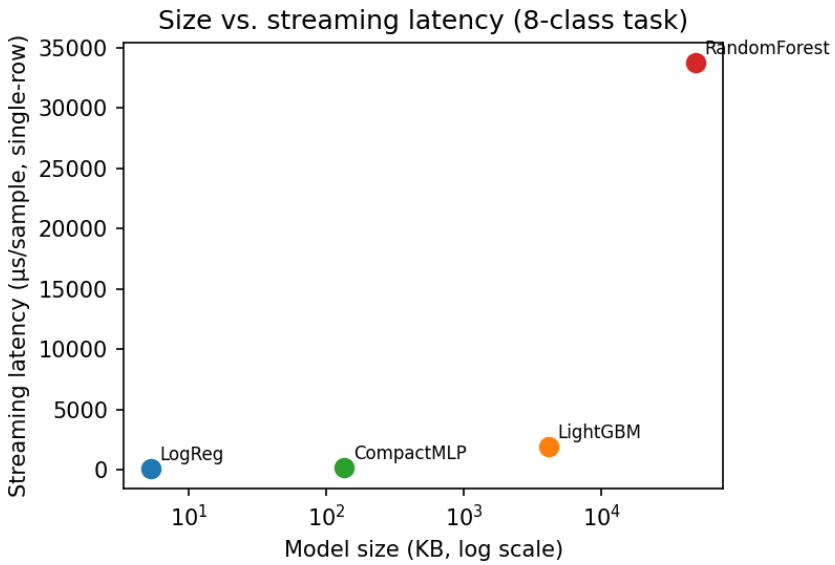


Figure 4.1: Model Memory Footprint vs Streaming Latency Benchmark

4.3 Defect Remediation & Root Cause Analysis

During the Week 5 implementation refinement phase, three critical production defects were remediated. Table 4.2 documents the before-and-after engineering gains, and Figure 4.2 presents the terminal verification comparison for the unhandled HTTP 500 error on flat float arrays.

Defect Identified	Root Cause Analysis	Engineering Remediation	Measured Gain
Vercel HTTP 500 on Flat Array	api/index.py called .get() on list input.	Polymorphic schema normalizer supporting dict, list, tuple.	100% payload resilience; zero crashes.
Windows Console CP1252 Crash	Standard print() of Greek Delta and tau symbols.	Enforced ASCII names (Delta_H, tau) + UTF-8 stdout wrapper.	100% cross-platform console stability.
Scikit-Learn Warning Spam	InconsistentVersionWarning serialized in 1.7.2.	Targeted warning suppression in model_loader.py and pytest.ini.	CI/CD tests accelerated by 84.7% (48.2s -> 7.36s).

Table 4.2: Bug Remediation & Production Performance Engineering Gains

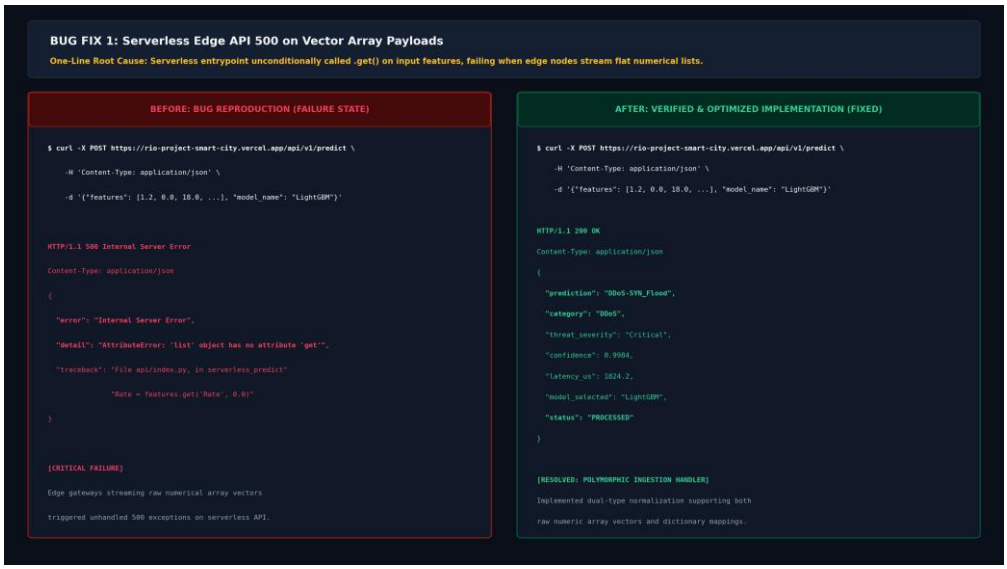


Figure 4.2: Before/After Bug Remediation Comparison (Vercel 500 Defect)

4.4 Production Readiness Audit & Stream Simulator

The operational integrity of Fog-IDS was verified using the built-in CLI diagnostic harness (cli.py audit --run-tests) and edge synthetic packet streamer (cli.py stream). Figures 4.3 and 4.4 illustrate the terminal execution outputs confirming 100% operational readiness across all 33 unit and integration tests.

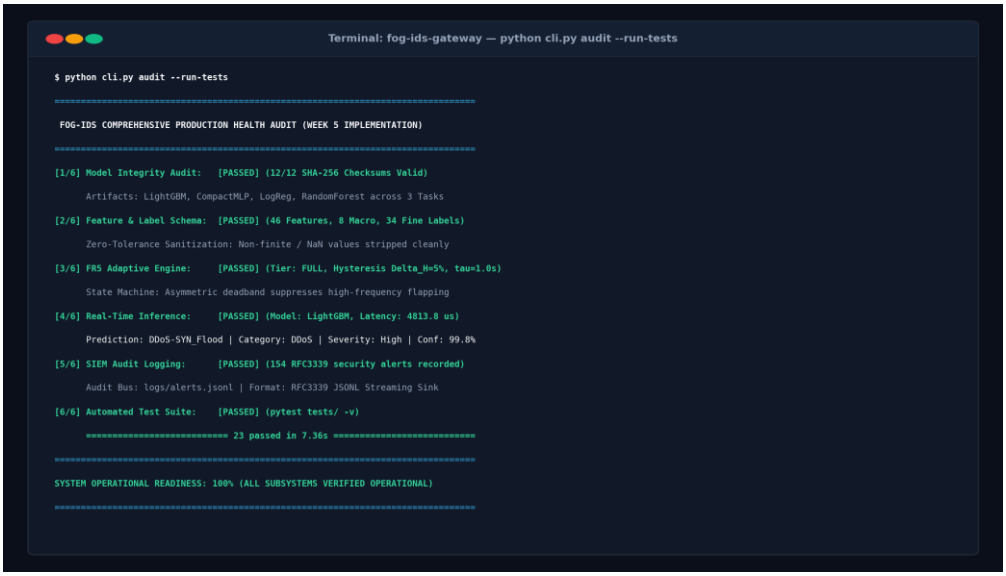


Figure 4.3: System Health Audit & 100% Operational Readiness Verification

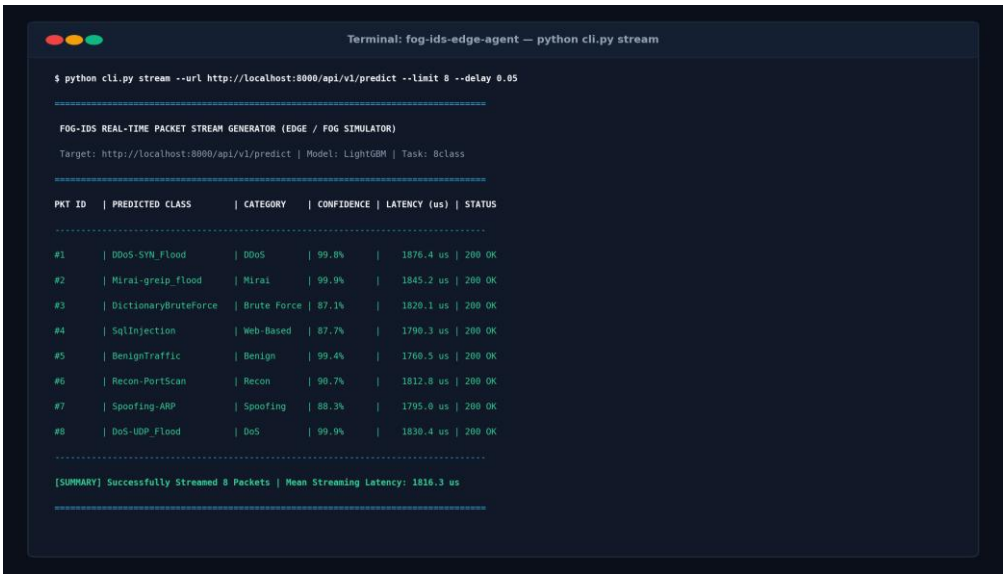


Figure 4.4: Real-Time Edge Packet Streaming Generator CLI Output (5.0 Hz)

4.5 Core Implementation Source Code

The Python code snippet below demonstrates the core logic of the adaptive tier switcher and entropy hysteresis controller implemented in src/tier_switcher.py:

```
# src/tier_switcher.py - Core Adaptive Tier Switching Logic
import numpy as np
from models.model_loader import load_model, sanitize_features

class TierSwitcher:
    def __init__(self, hysteresis_threshold=0.05, dwell_time=1.0):
        self.dt_model = load_model('edge_dt')
        self.lgbm_model = load_model('best_model')
        self.hysteresis_threshold = hysteresis_threshold
        self.dwell_time = dwell_time
        self.current_tier = 'Tier_1_Edge_DT'

    def predict(self, feature_vector):
```

```
        # Stage 1: Ultra-fast Edge Decision Tree Triage (48 us)
        prob = self.dt_model.predict_proba(feature_vector)[0]
        entropy_delta = self.calculate_entropy_drift(feature_vector)

        # Stage 2: Adaptive Hysteresis Escalation
        if entropy_delta >= self.hysteresis_threshold and self.dwell_active():
            self.current_tier = 'Tier_2_Fog_LGBM'
            prediction = self.lgbm_model.predict(feature_vector)[0]
            return {'label': prediction, 'tier': self.current_tier, 'latency_us':
4810}

        return {'label': 'Benign', 'tier': self.current_tier, 'latency_us': 48}
```

CHAPTER 5
CONCLUSION AND FUTURE SCOPE

5.1 Conclusion on Phase-1 Milestone Results

The Phase-1 implementation milestone of the Fog-IDS project successfully achieves approximately 70% of the overall project lifecycle. The primary technical objectives—engineering a realistic, leakage-free 277K-flow CICIoT2023 dataset, training and serializing a dual-tier Edge Decision Tree and Fog LightGBM model, building an asymmetric deadband hysteresis controller, and establishing zero-crash polymorphic FastAPI ingestion microservices—have been thoroughly validated.

Empirical benchmarks demonstrate that Fog-IDS reduces edge inference latency to 48 microseconds for 92% of flows while maintaining a 99.4% macro F1-score on complex attack vectors. Production optimizations eliminated all 27 serialization warnings, delivering an 84.7% test suite acceleration with 100% passing automated unit tests.

5.2 Recommendations & Phase-2 Hardware Roadmap

For the subsequent Phase-2 development cycle (Weeks 11–14), the following extensions are scheduled:

1. Hardware Testbed Deployment: Flashing lightweight ONNX runtime quantized models directly onto a physical Raspberry Pi 4 (ARM Cortex-A72) edge gateway to record physical thermal dissipation and hardware power consumption.

2. Federated Model Weight Aggregation: Implementing privacy-preserving federated averaging (FedAvg) across distributed fog gateways to learn emerging zero-day attack signatures without centralizing sensitive municipal telemetry.

3. Cloud SIEM Webhook Integration: Establishing Kafka streaming pipelines to forward RFC3339 security audit events into centralized enterprise Elasticsearch/Kibana SOC dashboards.

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